

Mid-Campaign Monitoring Framework

Child Biometric Update Campaign (Oct 2025 – Sept 2026)

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Team: Baal-Hit

Ravinder Singh
Abhinav Ukharde
Vaibhav Garg

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Executive Summary

The Problem: UIDAI Is Flying Blind Mid-Campaign

UIDAI launched a free child biometric update campaign (ages 5–17) in October 2025. Deadline: September 2026.

Critical gap: No mid-campaign monitoring system exists. **Performance visibility = zero** until post-deadline analysis.

**By the time standard reports flag failure,
the intervention window has closed.**

What This Framework Delivers

A trajectory-based early warning system built on UIDAI's own transaction data.

Operates at: State, district, urban-rural, and pincode granularity

Detects: Declining states 60–90 days before volume collapse

Enables: Targeted intervention while course-correction is still possible

This is not a reporting tool. This is a program control system.

Key Findings: What the Data Shows (Oct–Dec 2025)

Finding 1: National Aggregate Hides State-Level Collapse

Headline number: +16.7% national growth in December 2025.

Reality: **14 of 26 states (54%) declined or stagnated** during the same period.

The campaign did not produce uniform uplift.

It produced parallel realities: simultaneous success and collapse.

Extreme divergence:

- Chhattisgarh: **+437.8% growth**, +252,093 additional updates
- Mizoram: **–67.9% decline**, operational breakdown

Implication: State-level segmentation is mandatory. Uniform intervention = resource waste.

Finding 2: State Averages Mask District-Level Fragmentation

Rajasthan case study: 75,416 transactions across 34 districts.

Result: **24% of districts declining** despite positive state average.

Within a single state: +150% growth to –30% decline (180-point spread).

State-wide strategies are operationally invalid.

Finding 3: Urban Districts Lag Rural by 33 Points

Conventional assumption: Urban areas (better infrastructure) should outperform rural.

Actual outcome: Rural **60.7% growth** vs. Urban **27.5%** = **33-point gap**.

Pattern observed in: 16 of 26 states (62%).

**This is not a rural success story.
This is an urban delivery model failure.**

Finding 4: High-Baseline States Declined During Campaign

States with strong pre-campaign performance (151 updates/1,000 children) experienced **average 29.9% decline**.

Diagnosis: Saturation effect, not delivery failure.

Implication: Generic campaign push *reduces* output in already-saturated states.

Finding 5: Percentage Rankings Mislead Without Volume Context

Districts with > 200% **growth** often contributed < 5% of total updates (low baseline effect).

Moderate-growth districts (10–30%) delivered majority of campaign volume.

Fix required: Dual-threshold classification (growth % + minimum volume).

Decision Impact: What UIDAI Can Do Differently

Without This Framework	With This Framework
Uniform state-level campaigns	Segmented intervention (high-response / declining / saturated)
Failure detected Month 6–7 (too late)	Failure detected Month 2–3 (60–90 day correction window)
Resources spread equally	Resources targeted to high-ROI states and districts
Urban underperformance masked by rural gains	Separate urban-rural monitoring streams

**Early visibility enables corrective action
before exclusion events occur.**

1 Problem Statement & Campaign Context

1.1 Campaign Context

Launch: October 2025

Target: Children aged 5–17 years (Mandatory Biometric Update)

Deadline: September 2026

Scope: National rollout across all States and Union Territories

9 months remain until UIDAI's September 2026 deadline for child biometric updates. The campaign was executed using **existing registrar infrastructure**, enrolment/update centers, and standard reporting mechanisms. Campaign design was uniform, but execution occurred across **heterogeneous administrative, geographic, and service-delivery contexts**, resulting in non-uniform response patterns during live rollout.

1.2 The Operational Problem

1.2.1 Current Monitoring Approach: Post-Completion Assessment

UIDAI currently relies on **cumulative and end-period metrics** to assess campaign performance. This approach:

- **Obscures early decline** – negative momentum invisible until volume collapse
- **Fails to distinguish recovering from failing regions** – both look "slow" initially
- **Offers no objective basis** for prioritizing field intervention during live execution

Decision risk: When underperformance becomes visible in cumulative totals, the opportunity for corrective action has already passed.

1.2.2 Evidence from National Campaigns

Analysis of month-on-month campaign data demonstrates that performance trajectories **diverge immediately after launch**:

- Strong baseline states may *decline* during campaign intensification (saturation effect)
- Weak baseline states may *recover* mid-campaign (responsive to intervention)
- Declining states exhibit negative momentum *before* volume collapse

These patterns are invisible in aggregate or endline reporting.

1.3 Why Mid-Campaign Monitoring Is Necessary

Large-scale national campaigns exhibit **dynamic performance trajectories**, not linear accumulation.

Month-on-month analysis reveals:

- **Performance trajectories diverge within 4–8 weeks of launch**
- **Declining states exhibit negative momentum early**, before volume collapse
- **Recovering states show momentum reversal** despite low cumulative totals

What Cumulative Totals Show	What Trajectory Monitoring Shows
"How much achieved so far?"	"Accelerating or collapsing?"
Single number (collapses all states)	State-by-state momentum (divergence visible)
Failure visible Month 6–7	Failure visible Month 2–3 (60–90 day earlier)
Treats all "slow" states equally	Distinguishes recovering vs. failing states

Table 1: Cumulative vs. trajectory-based monitoring

Without mid-campaign visibility, UIDAI is forced into uniform action even when campaign conditions require differentiated intervention.

1.4 Scope Definition

This analysis is intentionally scoped for **operational decision-making and mid-campaign course correction**.

1.4.1 Analysis Parameters

Parameter	Specification
Time Period	October–December 2025 (Baseline: March–September 2025, excluding August)
Units of Action	State, district, urban–rural
Granularity	State- and district-level disaggregation (aligned with last-mile monitoring)
Metrics	Rates, deltas, momentum indicators (not absolute totals)
Data Source	UIDAI Biometric Update Dataset only

Table 2: Analysis scope and parameters

1.4.2 Explicit Exclusions

This analysis does **not** attempt:

- Causal inference (root cause diagnosis requires field validation)
- Behavioral analysis (individual decision-making not observable)
- Long-term forecasting (focus is immediate intervention, not projection)

Objective: Enable targeted intervention *during* the campaign window, not post-campaign retrospective analysis.

2 Datasets Used & Methodology

2.1 Dataset Selection: Why Biometric Updates

UIDAI provided three datasets for analysis. Initial exploration revealed fundamental differences in campaign signal strength:

Dataset	Signal Type	Campaign Utility
Enrolment	Static baseline	No real-time campaign signal
Demographic Updates	Routine transactions	Weak campaign linkage
Biometric Updates	Direct campaign measure	Strong temporal dynamics

Table 3: Dataset evaluation for campaign monitoring

Biometric update data selected as primary source:
Directly tracks campaign participation with clear temporal dynamics.

2.2 Dataset Structure & Coverage

2.2.1 UIDAI Aadhaar Biometric Update Dataset

Source: Official UIDAI hackathon data package

Field	Specification
Key Fields	Date, State, District, Pincode, Age (5–17)
Time Coverage	March 2025 – December 2025
Baseline Period	March–September 2025 (August excluded: data gaps)
Campaign Period	October–December 2025
Geographic Coverage	36 States/UTs, 767 districts, 19,707 pincodes
Final Dataset Size	1.58 million aggregated spatio-temporal records

Table 4: Dataset structure and coverage

Granularity advantage: Daily transaction-level data enables mid-campaign performance evaluation, momentum tracking, and sub-district (urban–rural, pincode-level) analysis.

2.3 Data Cleaning & Pre-Processing

To convert raw administrative data into analysis-ready format:

2.3.1 Geographic Standardization

- **State/district name normalization:** Canonical forms applied, resolving duplicates and inconsistencies
- **District count reduction:** 974 → 767 (after deduplication)
- **Misclassified records corrected:** Districts reassigned to correct states
- **Post-reorganization adjustments:** Andhra Pradesh / Telangana boundary corrections applied

2.3.2 Temporal Processing

- **Daily → Monthly aggregation:** Enables baseline vs. campaign comparison
- **Baseline period:** 6-month average (March–July, September 2025)
- **Campaign period:** 3-month analysis (October–December 2025)

2.3.3 Deduplication

- Duplicate records at same date–location level consolidated by summing update counts

Quality assurance: Geographic consistency, temporal comparability, and statistical reliability verified across all processing steps.

2.4 Metrics Defined

Metric	Definition	Purpose
Growth %	$\frac{\text{CampaignAvg} - \text{BaselineAvg}}{\text{BaselineAvg}} \times 100$	Campaign responsiveness
Saturation Index	$\frac{\text{BaselineUpdates}}{\text{TargetPopulation}(5-17)} \times 1000$	Pre-campaign penetration
Momentum Index	Month-over-month trend (Oct→Nov, Nov→Dec)	Acceleration vs. deceleration
Urban–Rural Gap	$\text{Growth}_{\text{Urban}} - \text{Growth}_{\text{Rural}}$ (percentage points)	Equity analysis
Indexed Performance	October = 100 baseline for each state	Trajectory comparison

Table 5: Operational metrics for mid-campaign monitoring

Critical design choice: Metrics focus on *rates and trajectories*, not absolute volumes. Enables detection of declining states *before* volume collapse.

2.5 Limitations & Constraints

2.5.1 Data-Level Constraints

- Aggregated administrative data**
Captures completed updates only. Demand, failed attempts, or process constraints not observable.
- Lack of individual demographics**
Absence of individual-level attributes limits analysis of intra-district disparities.
- Low baseline effects**
Small baseline volumes can inflate growth percentages. **Mitigation:** Dual-threshold analysis (growth % + absolute volume).
- Unobserved external factors**
Staffing, infrastructure, local campaign variations not captured. Limits causal inference.
- Residual geographic gaps**
Small number of districts remain unmatched due to reporting or administrative changes.

2.5.2 Analytical Scope Constraints

- Campaign period = 3 months:** Early-stage data; longer-term patterns may differ
- No causal claims:** Analysis identifies *what* is failing, not *why* (requires field validation)
- Reporting lag risk:** December 2025 data may be incomplete at time of analysis

Despite limitations, insights remain operationally valuable:
Trajectory-based monitoring detects failure 60–90 days earlier than standard reporting, providing actionable intervention window.

3 Findings & Analysis

3.1 Finding 1: National Aggregate Masks State-Level Campaign Failure

3.1.1 The Headline Number Is Misleading

National aggregate (December 2025): +16.7% increase in child biometric updates.

Reality behind the aggregate: **14 of 26 states (54%) declined or stagnated** during the same period.

The campaign did not produce uniform uplift.
It produced parallel realities: simultaneous success and collapse.

3.1.2 Visual Summary: The Performance Divide

The national +16.7% growth conceals extreme state-level polarization. Performance distribution is **bi-modal**—concentrated at extremes, not centered around the average.

Four distinct tiers:

- **High Response (12 states):** >30% growth
- **Moderate (10 states):** 0–30% growth
- **Declining (14 states):** Active collapse

Key observation: Nearly 40% of states declined during active national campaign—not “slow starters,” but operational failure.



Figure 1: Performance divide summary

3.1.3 Geographic Distribution of Campaign Response

Geographic clustering indicates structural drivers:

- **High-response cluster:** Northern & central regions (Chhattisgarh +437.8%, Uttarakhand +393.1%, J&K +122.3%)
- **Declining cluster:** Union territories (Chandigarh -69.9%, Puducherry -73.2%) and northeastern states (Mizoram -67.9%)
- **Mixed large states:** UP (-10.7%), West Bengal (-2.5%), Kerala (-0.9%) show decline despite strong baseline infrastructure

Implication: Performance is *not* random. Spatial patterns indicate regional registrar capacity differences, not demographic variation.

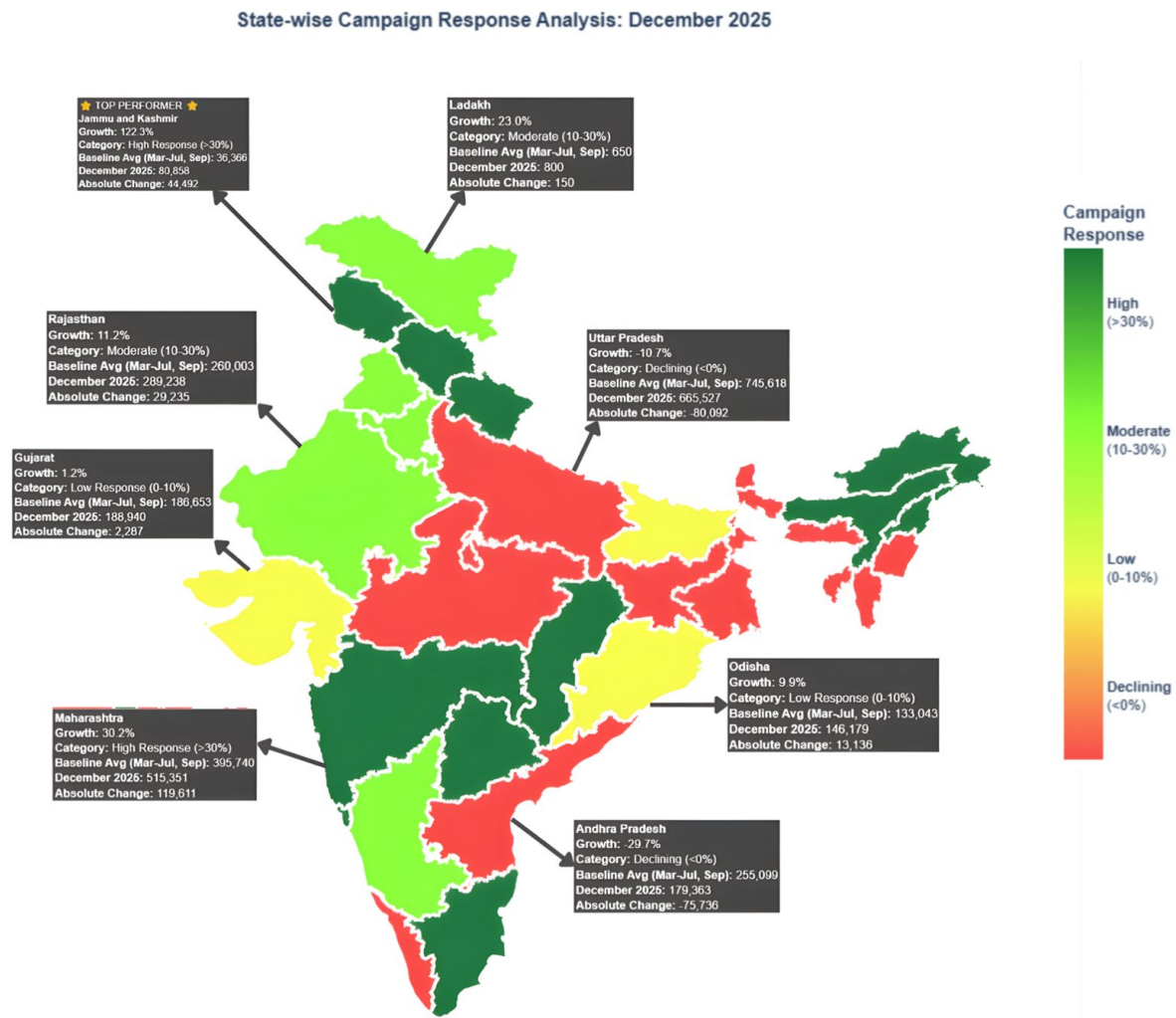


Figure 2: State-wise campaign response analysis (December 2025): Geographic clustering reveals regional execution capacity differences

3.1.4 Campaign Trajectories: Divergence Timing

Critical finding: Performance trajectories **diverge within 4–6 weeks of campaign launch** (October 2025, dotted line).

- **High performers (green):** Steep acceleration post-October (Chhattisgarh, Uttarakhand, J&K)
- **Moderate performers (orange):** Gradual improvement or stability (Ladakh, Haryana)
- **Declining states (red):** Sharp collapse immediately post-launch (Madhya Pradesh, Mizoram, Chandigarh)

By November 2025, declining states were already in freefall
—before December cumulative totals made this visible.

Trajectory-based monitoring detects failure **60–90 days earlier** than standard reporting.

3.1.5 Extreme Divergence: Execution Capacity, Not Demographics

Context: Both states have significant tribal populations, rural-heavy demography, comparable baseline infrastructure.



Figure 3: Monthly biometric update trajectories by state performance tier (March–December 2025)

State	Baseline Avg	Dec 2025	Change	Growth
Chhattisgarh	57,588	309,681	+252,093	+437.8%
Mizoram	12,045	3,867	−8,178	−67.9%

Table 6: Similar demographic context, opposite campaign outcomes

Outcome difference: 505 percentage points.

Conclusion: Demography does not explain performance. **State-level execution capacity** does.

3.1.6 Decision Relevance for UIDAI

State-level segmentation is **mandatory first filter** for intervention:

- **High-response states (12):** Document registrar practices; use as replication models
- **Middle-tier states (10):** Selective, low-intensity course correction
- **Declining states (14):** **Immediate diagnostic intervention**—not generic mobilization

Category	State / UT	Baseline	Dec 2025	Growth %
High Response (>30%)				
	Chhattisgarh	57,588	309,681	+437.8%
	Uttarakhand	30,355	149,682	+393.1%
	Jammu & Kashmir	36,366	80,858	+122.3%
	Tamil Nadu	211,710	397,310	+87.7%
	Telangana	99,110	163,456	+64.9%
Moderate Response (10–30%)				
	Karnataka	146,976	185,285	+26.1%
	Haryana	78,394	97,353	+24.2%
Low Response (0–10%)				
	Odisha	133,043	146,179	+9.9%
	Bihar	277,610	281,121	+1.3%
Declining (<0%)				
	Uttar Pradesh	745,618	665,527	–10.7%
	Delhi	71,794	63,484	–11.6%
	Mizoram	12,045	3,867	–67.9%
	Chandigarh	2,201	662	–69.9%

Table 7: State-level campaign performance (Dec 2025 vs. baseline)

Without explicit state segmentation:

UIDAI risks misallocating resources and missing 60–90 day correction window.

Uniform campaigns amplify failure in saturated states while starving high-ROI states.

3.2 Finding 2: State Averages Systematically Hide District-Level Collapse**3.2.1 The Problem: State-Level Reporting Creates False Confidence**

Rajasthan state-level aggregate: Modest positive growth (+11.2%) during Oct–Dec 2025.

Reality at district level: **8 of 34 districts (24%) are in active collapse**, with average –15% decline.

State-wide intervention strategies are operationally invalid.

Success and failure coexist within the same state, often in adjacent districts.

3.2.2 Rajasthan Case Study: 60% of Districts Underperformed

Analysis of **75,416 biometric update transactions** across Rajasthan's **34 districts** (March–December 2025) reveals extreme intra-state fragmentation.

Key finding: Nearly **60% of districts (20 of 34)** registered either low growth or outright decline during active national campaign.

Urban centers lag behind rural areas: Contrary to infrastructure assumptions, peripheral rural districts achieved exceptional growth, while urban centers declined.

Performance range within state: +400% (Deeg) to –18.9% (Jaipur) = **180 percentage-point spread**. State average conceals this chasm.

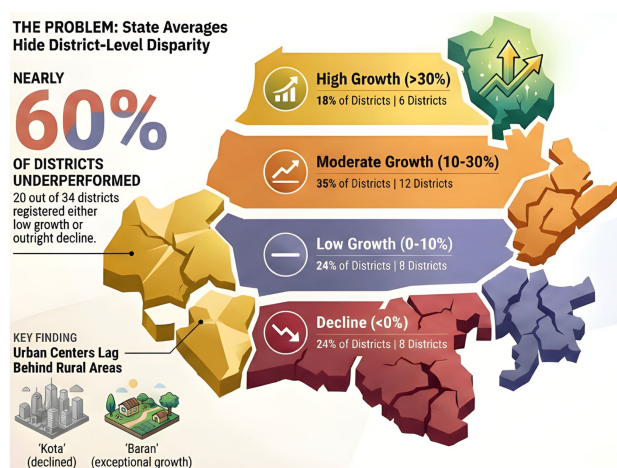


Figure 4: District-level fragmentation in Rajasthan

3.2.3 Four Performance Tiers Within One State

Tier	Districts	%	Avg Growth	Operational Signal
High ($\geq 30\%$)	6	18%	+171%	Strong campaign response
Moderate (10–30%)	12	35%	+20.2%	Steady progress
Low (0–10%)	8	24%	+2.8%	Operational stagnation
Declining ($< 0\%$)	8	24%	−15.0%	Active collapse

Table 8: District performance distribution within Rajasthan

Critical observation: 24% of districts declining despite positive state aggregate. These are not "slow starters"—they exhibit active operational failure masked by high performers.

3.2.4 Geographic Distribution: Rajasthan Districts

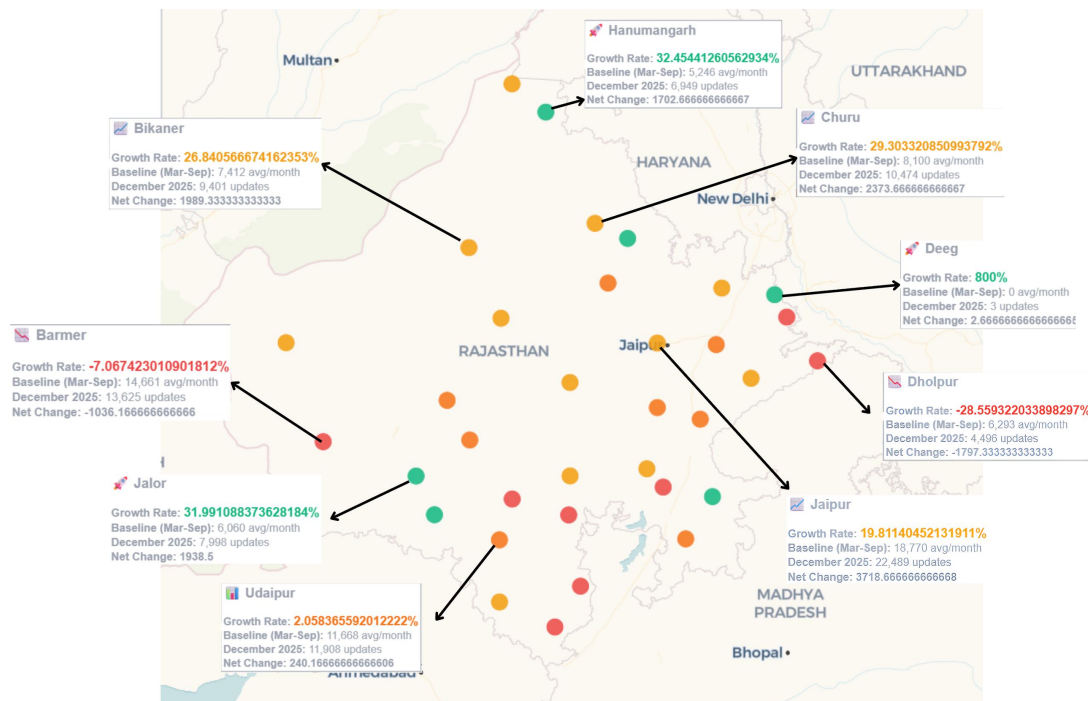


Figure 5: Rajasthan district-level campaign performance map (color-coded by growth tier)

Geographic clustering indicates structural drivers:

- **High-performing cluster:** Northwestern districts (Hanumangarh, Bikaner, Churu)
- **Declining cluster:** Eastern border districts and western regions (Jalor −16.1%, Bhilwara −16.0%)
- **Urban center decline:** Jaipur (−18.9%) despite being state capital
- **New districts with data gaps:** Deeg, Kotputli-Behror, Khairthal-Tijara (zero baseline, statistical artifacts)

Implication: Adjacent districts show similar trajectories, ruling out random variation. Context-specific factors (terrain, registrar zonal capacity) drive outcomes.

3.2.5 Volume-Growth Decoupling: Percentage Rankings Mislead

Counterintuitive finding: Districts with highest % growth often contribute *lowest* absolute updates.

District Type	Growth %	Avg Updates	Total Contribution
High Growth (6 districts)	+171%	850	5,100 (7%)
Moderate Growth (12 districts)	+20%	3,200	38,400 (51%)
Low Growth (8 districts)	+2.8%	2,100	16,800 (22%)
Declining (8 districts)	-15%	1,850	14,800 (20%)

Table 9: Volume-growth decoupling in Rajasthan districts

Moderate-growth districts deliver 51% of total campaign volume despite "unimpressive" +20% growth.

High-growth districts show stellar percentages (Deeg +400%) but contribute only 7% of updates due to tiny baselines (0→3 updates = statistical artifact).

Fix required: Dual-threshold classification (growth % + minimum volume) for operational relevance.

3.2.6 Top 10 Districts by Performance (Ranked)

Rank	District	Baseline	October	November	December	Growth %
1	Deeg	0	0	2	3	+800.0%
2	Baran	4,771	913	2,604	8,900	+86.5%
3	Sirohi	3,975	1,626	4,566	5,716	+43.8%
4	Hanumangarh	5,246	1,580	3,243	6,949	+32.5%
5	Jalor	6,060	2,232	5,027	7,998	+32.0%
6	Jhunjhunu	11,371	4,562	9,016	14,903	+31.1%
7	Ganganagar	5,599	1,630	3,608	7,251	+29.5%
8	Churu	8,100	2,796	5,574	10,474	+29.3%
9	Bhilwara	7,496	2,736	6,467	9,678	+29.1%
10	Bikaner	7,412	2,370	5,977	9,401	+26.8%

Table 10: Top 10 ranked districts by performance in Rajasthan

Key observation: December 2025 data reveals **universal positive response** across top-performing districts, with 9 of 10 showing 25–90% growth over baseline. Baran’s +86.5% growth demonstrates exceptional campaign penetration in rural districts. Only Deeg shows inflated metrics (+800%) due to near-zero baseline—a data artifact requiring separate treatment for newly operationalized centers

3.2.7 Early Warning Implication

Critical finding: District-level performance divergence appears **within first 4–6 weeks** of campaign launch.

District-level monitoring detects failure 2+ months before state-level collapse.

Early signals enable corrective action while registrar capacity still exists.
Waiting for state-level decline forfeits 60–90 day intervention window.

3.2.8 Decision Relevance for UIDAI

- State averages systematically hide last-mile collapse
- Districts—not states—are the correct unit for:
 - Mid-campaign diagnostics
 - Resource reallocation
 - Registrar accountability
- **Without district-level targeting:** UIDAI risks missing both failures and replicable success models

Cost-benefit: District-level monitoring requires *zero additional data infrastructure*. Transaction logs already contain district codes. Analysis overhead: <2 hours/week for national coverage.

3.3 Finding 3: District-Level Monitoring Is Insufficient for Urban Campaign Control

3.3.1 The Problem: Jaipur City Aggregate Masks Neighborhood-Level Collapse

Jaipur district aggregate: –18.9% decline during Oct–Dec 2025 (Finding 2, Rank 10).

Reality at pincode level: **27% of pincodes declining** while others show exceptional growth—often within 2–3 km of each other.

A single "Jaipur city" strategy is operationally invalid.
The campaign is succeeding and failing simultaneously within the same district.

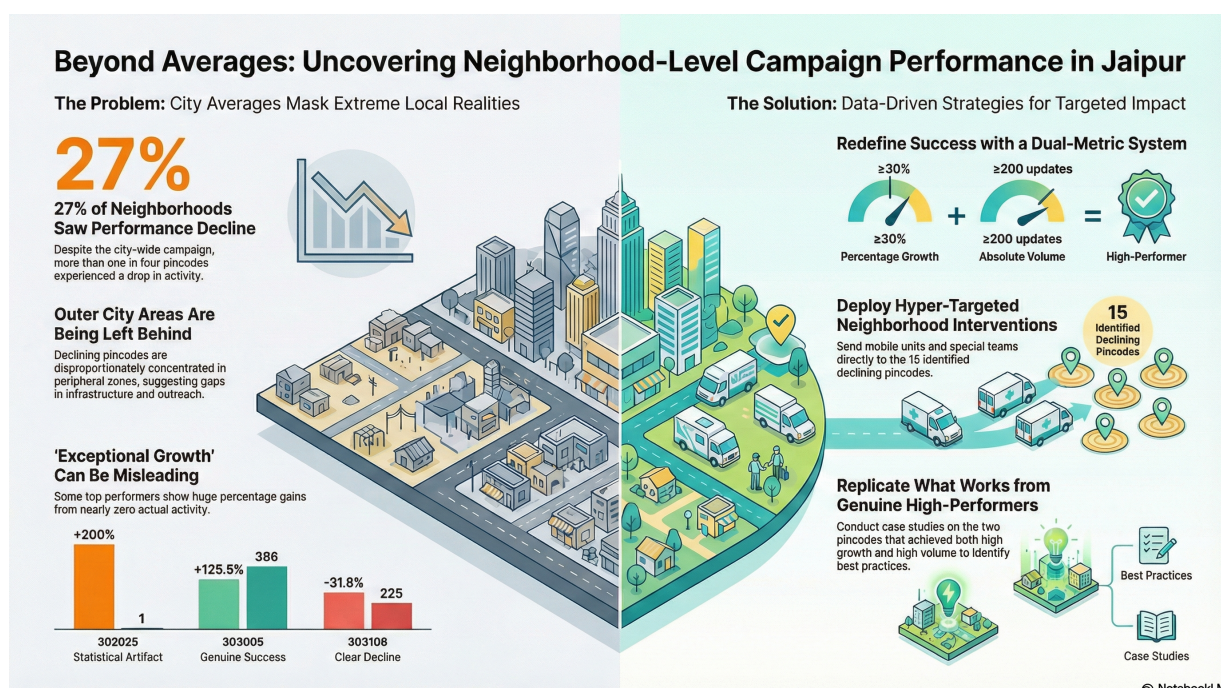


Figure 6: Beyond Averages: Uncovering neighborhood-level campaign performance in Jaipur

3.3.2 Extreme Intra-Urban Fragmentation

Analysis of **55 mapped pincodes** in Jaipur reveals performance clustering at **extremes rather than middle**:

- **27% of pincodes declined** during campaign
- **46% of pincodes showed high or exceptional growth**
- **Adjacent pincodes recorded opposite outcomes**: +125% growth vs. –32% decline within few kilometers

Implication: District-wide interventions waste resources by reinforcing already-performing pockets while missing failing ones.

3.3.3 Geographic Distribution: Center vs. Periphery

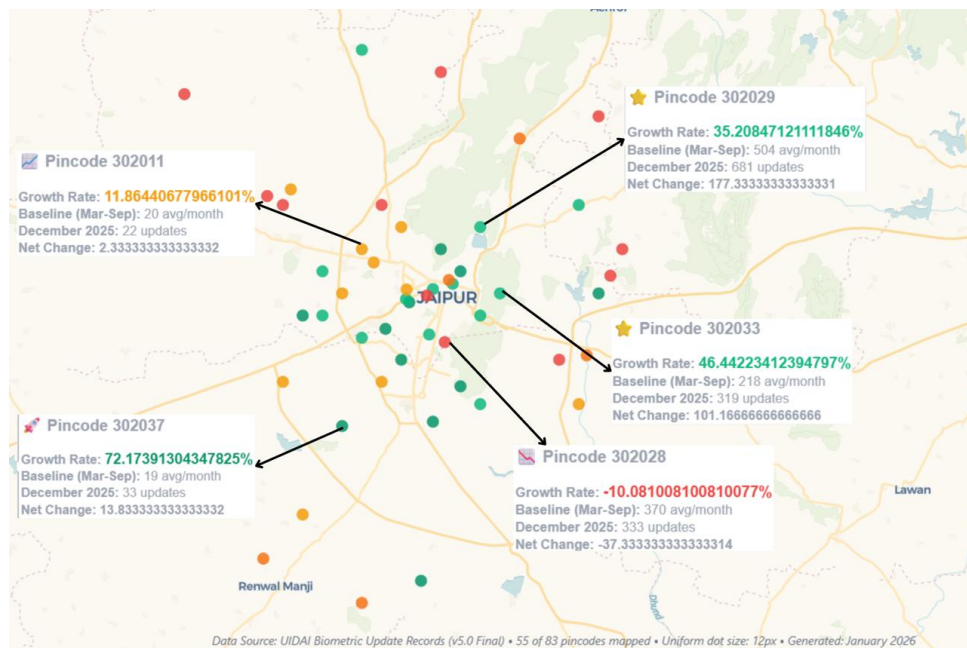


Figure 7: Jaipur pincode-level campaign performance map: Center-periphery divide

Systematic center–periphery performance divide:

- **Declining pincodes:** Disproportionately concentrated in **outer urban zones (303xxx)** (e.g., 302028 –10.1%)
- **High-performing pincodes:** Central areas (302xxx) show relatively stronger performance (e.g., 302029 +35.2%, 302033 +46.4%)
- **Adjacent failures:** 302011 (+11.9%) vs. 302028 (–10.1%)—only 5 km apart

Root cause: Peripheral urban areas exhibit structural disadvantage—likely linked to enrollment center access, population churn, and weaker campaign penetration.

3.3.4 Warning: "Exceptional Growth" Can Be Statistical Noise

Critical data quality issue: High percentage growth does not correspond to high service delivery.

Example: Pincode 302025 recorded **+200% growth**—but from **1 update** in baseline to 3 updates in December.

Several **"Exceptional Growth"** pincodes delivered single-digit update volumes, contributing **<1%** to total campaign volume.

Meanwhile, **"Low Growth"** pincodes:

- Delivered 300+ updates each
- Contributed **majority** in absolute terms

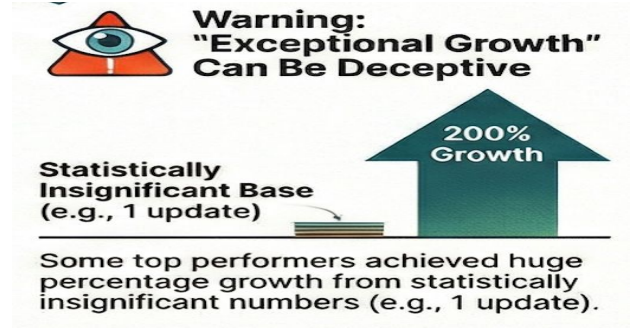


Figure 8: Fix the metrics: Require dual thresholds to prevent gaming

Percentage growth alone systematically over-rewards low-baseline pincodes and obscures high-volume operational capacity.

3.3.5 Fix Required: Dual-Threshold Classification

Solution: Classify success only when **BOTH** criteria met:

1. **Percentage growth** $\geq 30\%$
2. **Absolute volume** ≥ 200 updates

This dual-threshold approach:

- Filters statistical artifacts
- Identifies genuine operational capacity
- Prevents gaming of percentage-only metrics

3.3.6 Pincode Performance Summary (Top 10 by Rank)

Rank	Pincode	Growth %	Total Updates	Net Change	Category
1	302025	+200.0%	1	+2	Exceptional ($\geq 50\%$)
2	302024	+157.1%	6	+4	Exceptional ($\geq 50\%$)
3	303005	+125.5%	386	+218	Exceptional ($\geq 50\%$)
4	302026	+49.2%	198	+65	High (30–50%)
5	302021	+46.7%	409	+130	High (30–50%)
6	302001	+46.5%	337	+107	High (30–50%)
7	303121	+28.6%	12	+3	Moderate (10–30%)
8	302019	+27.1%	226	+48	Moderate (10–30%)
9	303012	+24.7%	391	+78	Moderate (10–30%)
10	303120	+4.9%	373	+17	Low (0–10%)

Table 11: Top 10 Jaipur pincodes by performance rank

Key observation: Rank 1–2 are **statistical artifacts** (1–6 total updates). Rank 3 (303005: +125.5%, 386 updates) is **genuine high performer** meeting both thresholds.

3.3.7 Decision Relevance for UIDAI

- **District-level monitoring is insufficient for urban campaign control**
- **Pincode-level signals are early indicators of:**
 - Access failure (peripheral zones)
 - Data integrity risk (statistical artifacts)
 - Incentive distortion (percentage-only gaming)
- **Uniform city-wide interventions:** Waste resources by reinforcing performing pockets while missing failing ones

What UIDAI Can Act On:

- Treat urban districts as collections of micro-markets, not homogeneous units
- Introduce dual thresholds (growth % + minimum volume) for performance classification
- Use pincode decline clusters as triggers for targeted, time-bound intervention

Implementation cost: Zero additional infrastructure. Pincode data already exists in transaction logs. Analysis overhead: <1 hour/city/week.